Nonresponse and Measurement Error

STA304: Surveys, Sampling, and Observational Data

Emily Somerset, Thursday June 11, 2026

Housekeeping

- Last lecture today
- Final Exam on Tuesday, June 23rd.
- Course evaluation form
- Final Exam page + formula sheet + practice problems will be posted this Friday.
- Learning Activity 9 due Friday June 12 at 12pm.
- Learning Activity 10 due Tuesday June 16 at 11:59pm.

Survey Sam & Obs Data
STA304H1-F-LEC0101

Students

https://go.blueja.io/mZZ6LU4YWU6SOS5_tYY4tQ

Learning outcomes

- [ ] Describe common sources of nonresponse and measurement error in surveys.
- [ ] Explain how nonresponse and measurement error can affect the accuracy of survey estimates.
- [ ] Construct nonresponse-adjusted survey weights using weighting class adjustments, post-stratification, and raking.
- [ ] Explain how auxiliary information can be used to reduce nonresponse bias.
- [ ] Describe the purpose of imputation and explain how imputation can be used to address missing data.
- [ ] Describe common strategies for reducing measurement error in survey design and data collection.

Last time

1. We talked about response propensities

$$\phi_i = P(R_i = 1) = P(\text{unit } i \text{ responds})$$

2. If we knew ϕ_i, an unbiased vestimator for the total would be:

$$\hat{t}_\phi = \sum_{i=1}^{N} I_i R_i \frac{y_i}{\pi_i \phi_i} = \sum_{i \in \mathcal{R}} \frac{y_i}{\pi_i \phi_i} \leftarrow$$

upweight it by # of non-respondents it represents, and by non-sampled pop it rep.

3. We discussed using auxiliary information in order to estimate $\hat{\phi}$

4. Our ability to come up with a good estimate of $\hat{\phi}_i$ depends on the missing mechanism (the relationship between ϕ_i and the auxiliary information we have available to us).

Today

We will talk about 3 methods to adjust for nonresponse

1. Use information from the $\mathbf{x}$ variables to estimate the probability that each sampled unit i responds (response propensity). Use these predictions to **adjust weights for nonresponse.**

2. Use **post stratification** to adjust weights of the sample so that estimates of population total or mean for the $\mathbf{x}$ variables agree with known population totals.

3. Use information from the $\mathbf{x}$ variables to predict the value of y_i. This is known as **imputation.**

8.5 Adjusting Weights for Nonresponse

Estimator

- We've seen how weights can be used in calculating estimates under various sampling designs
- $\hat{t} = \sum_{i \in S} \omega_i y_i$ with $\omega_i = \dfrac{1}{\pi_i} = \dfrac{1}{P(I_i = 1)}$

8.5 Adjusting Weights for Nonresponse

Estimator

Examples:

- For stratified sampling, these weights were $\omega_i = N_h/n_h$ if unit i is in stratum h

- For SRS, these weights were $\omega_i = N/n$

- For unequal probability, these weights were $\omega_i = \dfrac{1}{\pi_i} = \dfrac{1}{\psi_i}$

8.5 Adjusting Weights for Nonresponse

Estimator

- Weights can also be used to adjust for nonresponse

- If we knew ϕ_i we could assign a weight of $\dfrac{1}{\pi_i \phi_i}$ to the value of y_i

$$\hat{t}_{\phi} = \sum_{i \in \mathcal{R}} \frac{y_i}{\pi_i \phi_i}$$

- Or we can use an estimator for ϕ_i, $\hat{\phi}_i$ and plug it in:

$$\hat{t}_{\hat{\phi}} = \sum_{i \in \mathcal{R}} \frac{y_i}{\pi_i \hat{\phi}_i}$$

8.5 Adjusting Weights for Nonresponse

Estimating propensity scores

- For the two next propensity score estimation methods, we assume the auxiliary information we have is of form (A) or (B)

- We discuss 2 methods:

    1. Weighting Class Adjustments

    2. Regression models

8.4 Response propensities

Auxiliary information for treating nonresponse

- This auxiliary information can come in different forms.

    A. Known for every unit in the sampling frame. For example, a list of registered voters, used as sampling frame, may contain voter's age, sex, history of participating in previous elections, etc…

    B. Known for the sampled units. For example, you might be interested in income but collected other information on the units like age, job title, highest level of education, etc..

    C. Population means or totals from an external data source. For example, demographics from census surveys or population registry.

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- Recall, sampling weights ω_i are interpreted as the number of units in the population represented by unit i in the sample.

- Taken together, the sample represents all N members in the population.

- And thus: $N \approx \sum_{i \in S} \omega_i$

e.g SRS , $\omega_i = \frac{N}{n} \quad \forall i \in S$

$$\sum_{i \in S} \omega_i = \sum_{i \in S} \frac{N}{n} = \left(\frac{N}{n}\right) \sum_{i \in S} (1) = \boxed{N}$$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- With nonresponse, we don't have any data on some selected units
- Weighting class adjustments (the topic of today) shifts the weight of those who don't respond to the respondents.
- Call this new weight $\tilde{\omega}_i$
- And still we have $N \approx \sum_{i \in \mathscr{R}} \tilde{\omega}_i$

$\mathscr{R}$: set of respondent

- How do we create these weights?

8.5 Adjusting Weights for Nonresponse

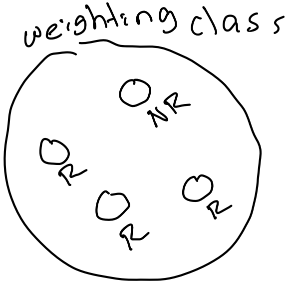
Estimating response propensities: weighting class adjustment

- Shifts the weight of those who don’t respond to the respondents.
- Call this new weight $\tilde{\omega}_i$
- To do this we need to decide how to shift these weights. To what responding unit do we give the weight of a non-responding unit?

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- Consider the idea of dividing our sample into groups.
- We call these groups: weighting classes.
- If a non-respondent belong to a weighting class, we will shift all of its weight to the respondents in that class.



8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- **Step 1: Divide the units into weighting classes**
  - Divide the units in the selected sample S among C weighting classes so that each unit belongs to 1 class.
  - These classes are formed using auxiliary information known for all units in S.
  - Let $x_{ic} = 1$ if unit i is in class c and 0 otherwise.
  - Example: Separating the sample into different age groups.

e.g. age groups

≤ 19, $19-30$, >30
(1) (2) (3)

$x_{emily,3} = 1$

$x_{emily,2} = 0$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- **Step 2: Redistribute the weights from nonrespondents in class c to respondents in class c**
  - It is assumed that people in the same class will have similar outcome values y_i

$$\tilde{\omega}_i = \omega_i \left(\frac{\text{sum of weights for selected sample in c}}{\text{sum of weights for respondents in class c}} \right)$$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- **Step 2: Redistribute the weights from nonrespondents in class c to respondents in class c**
  - It is assumed that people in the same class will have similar outcome values y_i

$$\tilde{\omega}_i = \omega_i \frac{\text{sum of weights for selected sample in c}}{\text{sum of weights for respondents in class c}}$$

$$\frac{1}{\pi_i}$$

$$\frac{1}{\hat{\phi}_c}$$

$\hat{\phi}_c = \dfrac{\text{sum of weights resp in c}}{\text{sum of weights selected in c}}$

proportion of resp in class c

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- **Step 2: Redistribute the weights from nonrespondents in class c to respondents in class c**

$$\tilde{\omega}_i = \omega_i \frac{\text{sum of weights for selected sample in c}}{\text{sum of weights for respondents in class c}}$$

Example 1: SRS of 20 people and 10 people within age group 20-25 with 2 non-respondents and 10 people within age group 25-30 with 5 non-respondents. Population size 100.

For weight class 1: 20-25

$$\tilde{\omega}_i = \omega_i \left[\frac{\text{sum of selected}}{\text{sum of resp}} \text{" "}\right] = \frac{100}{20} \times \left[\frac{10 \times \left(\frac{100}{\cancel{20}}\right)}{8 \times \left(\frac{100}{\cancel{20}}\right)}\right] = 5 \times \left(\frac{10}{8}\right) = 6.25$$

For class 2: 25-30

$$\tilde{\omega}_i = \frac{100}{20} \times \left(\frac{10}{5}\right) = 5 \times 2 = 10$$

→ both selected sample size in each weighting class are the same ($n_1 = 10$, $n_2 = 10$)

but we have a lot more missing in the older age group

⇒ so we can see why they have a higher weight.

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- **Step 2: Redistribute the weights from nonrespondents in class c to respondents in class c**

After this step, the weights for the nonrespondents in class c have been transferred to the respondents in class c.

A respondent in class C now has its original weight ω_i and additional weight for non-respondents.

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- **Step 3: Double check calculations**

Check that $\sum_{i \in \mathcal{R}} \tilde{\omega}_i = \sum_{i \in S} \omega_i$

This means all the weight has been redistributed to the respondents.

Exercise: Check this for the example 1.

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- **Step 3: Double check calculations**

Check that $\sum_{i \in \mathscr{R}} x_{ic} \tilde{\omega}_i = \sum_{i \in S} x_{ic} \omega_i$

This means all the weight has been redistributed to the respondents in class c.

Exercise: Check this for the example 1.

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- This method relies on $P(R_i = 1) = \phi_c$ for every unit in weighting class c
- That is, the missing data mechanism is MCAR WITHIN a class.
- It assumes MAR more generally.

$$\hat{\phi}_c = \frac{\text{sum of weights for respondents in class c}}{\text{sum of weights for selected sample in c}}$$

← it's like a proportion est. of responses

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- Generally, our estimators for total and mean are:

$$\hat{t}_{\hat{\phi}} = \sum_{i \in \mathscr{R}} \frac{y_i}{\pi_i \hat{\phi}_i} = \sum_{i \in \mathscr{R}} \tilde{\omega}_i y_i$$

$$\hat{\bar{y}}_{\hat{\phi}} = \frac{\hat{t}_{\hat{\phi}}}{\sum_{i \in \mathscr{R}} \tilde{\omega}_i} = \frac{\sum_{i \in \mathscr{R}} \tilde{\omega}_i y_i}{\sum_{i \in \mathscr{R}} \tilde{\omega}_i}$$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

Example 1: SRS of 20 people from a population of size 100. There are 8 people within age group 20-25 with 2 non-respondents and 12 people within age group 25-30 with 5 non-respondents.

The outcome of interest is income.

The mean of the respondents in age group 20-25 is $\bar{y}_{1R} = 70{,}000$ and $\bar{y}_{2R} = 90{,}000$

R_1 : respondents in 20-25

R_2 : respondents in 25-30

Estimate the population mean and total using weighting class adjustments.

20-25:

$$\tilde{w}_i = w_i \times \left(\frac{8\times(\cancel{N/n})}{6\times(\cancel{N/n})}\right) = \frac{100}{20}\times\frac{8}{6} = 5\times 4$$

$$= 6.67$$

25-30

$$\tilde{w}_i = \frac{100}{20}\left(\frac{12}{7}\right) = 5\times\left(\frac{12}{7}\right) = 8.57$$

$$\hat{t}_{\phi} = \sum_{i\in R}\tilde{w}_i y_i = \sum_{i\in R_1}\tilde{w}_i y_i + \sum_{i\in R_2}\tilde{w}_i y_i$$

$$= 5\times 8\times\frac{1}{6}\sum_{i\in R_1} y_i + 5\times 12\times\frac{1}{7}\sum_{i\in R_2} y_i$$

$$= 5\times 8\times 70{,}000 + 5\times 12\times 90{,}000$$

$= 8{,}200{,}000$ $\boxed{\hat{\bar{y}}_{\hat{\phi}} = 82{,}000}$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

Example 2:

TABLE 8.2
Illustration of Weighting Class Adjustment Factors

	Age					Total
	15–24	25–34	35–44	45–64	65+	
Sample size	202	220	180	195	203	1000
Respondents	124	187	162	187	203	863
Sum of weights for sample	30,322	33,013	27,046	29,272	30,451	150,104
Sum of weights for respondents	18,693	28,143	24,371	28,138	30,451	
$\hat{\phi}_c$	0.6165	0.8525	0.9011	0.9613	1.0000	
Weight factor	1.622	1.173	1.110	1.040	1.000	

R_2 : response set in class 25-34

S_2: sampled set in 25-34

Check that $\sum_{i \in \mathcal{R}} x_{ic}\tilde{\omega}_i = \sum_{i \in S} x_{ic}\omega_i$ for weighting class 25-34

25-34 $\quad \sum_{i \in S} x_{ic}\omega_i = \sum_{i \in S_2} \omega_i = 33{,}013$

25-34 $\quad \sum_{i \in R_2} \tilde{\omega}_i = \sum_{i \in R_2} \omega_i \times \frac{1}{\hat{\phi}_2}$

$= \sum_{i \in R_2} \omega_i \left(\frac{\text{sum of sample}}{\text{sum of resp}}\right)$ $\quad$ // 1.173

$= \sum_{i \in R_2} \omega_i \left(\frac{33013}{28143}\right) = \left(\frac{33013}{28143}\right) \sum_{i \in R_2} \omega_i$

$\boxed{= 33013}$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

How should we choose the weighting classes?

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

$$\hat{t}_{\hat{\phi}} = \sum_{i \in \mathscr{R}} \frac{y_i}{\pi_i \hat{\phi}_i} = \sum_{i \in \mathscr{R}_1} \frac{y_i}{\pi_i \hat{\phi}_1} + \ldots + \sum_{i \in \mathscr{R}_C} \frac{y_i}{\pi_i \hat{\phi}_C}$$

- Weighting class adjustments produce approximately unbiased estimators

  1. When ϕ_i in a class c are all the same: $\phi_i \approx \phi_j$ in class c.

⇝ remember we are a estimating a single $\hat{\phi}_c$ for a class

⇝ so it makes sence to use a single $\hat{\phi}_c$ to weight the respondents

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

$$\hat{t}_{\hat{\phi}} = \sum_{i \in \mathscr{R}} \frac{y_i}{\pi_i \hat{\phi}_i} = \sum_{i \in \mathscr{R}_1} \frac{y_i}{\pi_i \hat{\phi}_1} + \ldots + \sum_{i \in \mathscr{R}_C} \frac{y_i}{\pi_i \hat{\phi}_C}$$

- Weighting class adjustments produce approximately unbiased estimators

2. When y_i in a class are all the same.

Unit	True weight	Weight used	y
A	8	6	50
B	2	4	52

doesn't matter if we use the wrong weights

if we expect resp. and non-respondents should have similar y_i's

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

$$\hat{t}_{\hat{\phi}} = \sum_{i \in \mathscr{R}} \frac{y_i}{\pi_i \hat{\phi}_i} = \sum_{i \in \mathscr{R}_1} \frac{y_i}{\pi_i \hat{\phi}_1} + \ldots + \sum_{i \in \mathscr{R}_C} \frac{y_i}{\pi_i \hat{\phi}_C}$$

- Weighting class adjustments produce approximately unbiased estimators

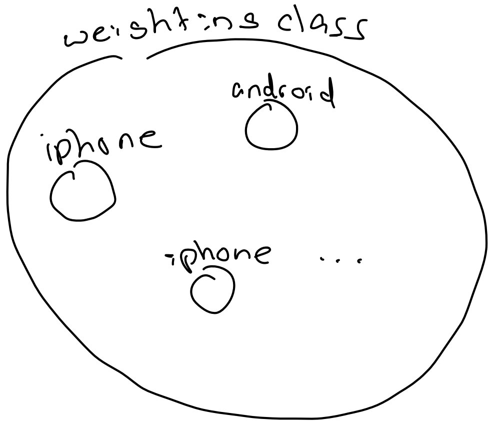
    3. When y_i is uncorrelated to ϕ_i in a class

if missingness in y is completely uncorrelated to the outcome

y: hours of exercise per week and x: android vs iphone

my survey is glitchy on iphone

→ clearly phone type is uncorrelated with exercise



phonetype	hours exercise	ϕ_i
iphone	4	0.25
iphone	6	0.25
android	4	0.75
android	6	0.75

→ exercise hours for android users is still a representative sample of the population in that class

e.g

→ the mean estimator of the respondents in that class will be representative of the population mean of that class

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: weighting class adjustment

- It's not likely that you can create weighting classes so that one of these conditions will hold exactly.
- Good to try to create weighting classes keeping this idea in mind.

Example: You want to estimate the population proportion of smoking. You suspect women will be less likely to respond than men and younger people will be less likely to respond than older people

Can create 4 weighting classes, women + younger, women + older, etc..
and assume similar propensity to answer in each class.

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: regression models

- The weighting class adjustment model is commonly adopted to estimate the response propensities ϕ_i

$r = [1, 0, 1, 1, 1, \ldots]$

- But a regression model could also be used!

1 = respond

- That is we model the outcome R_i as a function of $\mathbf{x}_i$

0 = no response

- In particular, a logistic regression model or regression trees are very useful.

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: regression models

Logistic regression model:

$$R_i \sim \text{Bernoulli}(\phi_i)$$

logit

$$\log\left(\frac{\phi_i}{1-\phi_i}\right) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \ldots$$

$\phi_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots$

logit parametrizes

ϕ_i so it takes

$(-\infty, \infty)$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: regression models

Logistic regression model:

$$R_i \sim \text{Bernoulli}(\phi_i)$$

$$\log\left(\frac{\phi_i}{1-\phi_i}\right) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \ldots$$

$$\frac{\exp(x_i \beta)}{1 + \exp(x_i \beta)} \in [0, 1]$$

Some other ways to write the second line:

$$\text{logit}(\phi_i) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \ldots$$

$$\phi_i = \frac{\exp(\mathbf{x}_i \beta)}{1 + \exp(\mathbf{x}_i \beta)} \qquad \phi_i = \text{logit}^{-1}(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \ldots) = \underbrace{\text{logit}}^{-1}(\mathbf{x}_i \beta)$$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: regression models

Logistic regression model:

statistical model

$$R_i \sim \text{Bernoulli}(\phi_i)$$

$$\log\left(\frac{\phi_i}{1-\phi_i}\right) = \underbrace{\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \ldots}_{\vec{x}_i \beta}$$

→ data we give R is r_i, x_{1i}, x_{2i}

⇒ when you fit it $\hat{\beta}_0, \hat{\beta}_1, \ldots$

- Once this model is fit, you can obtain estimates of ϕ_i directly from model

$$\log\left(\frac{\hat{\phi}_i}{1-\hat{\phi}_i}\right) = \vec{x}_i \hat{\beta}$$

$$\frac{\hat{\phi}_i}{1-\hat{\phi}_i} = \exp(\vec{x}_i \hat{\beta})$$

$$\hat{\phi}_i = \frac{\exp(\mathbf{x}_i\hat{\beta})}{1+\exp(\mathbf{x}_i\hat{\beta})} \longrightarrow \text{Then } \tilde{\omega}_i = \omega_i \frac{1}{\hat{\phi}_i}$$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: regression models

Logistic regression model:

$$R_i \sim \text{Bernoulli}(\phi_i)$$

$$\log\left(\frac{\phi_i}{1-\phi_i}\right) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \ldots$$

- You can then calibrate the weights so that they sum to N

$$\tilde{\omega}_i = \omega_i \frac{1}{\hat{\phi}_i} \qquad \omega_i^* = \tilde{\omega}_i \times \frac{N}{\sum_{i \in \mathscr{R}} \tilde{\omega}_i}$$

8.5 Adjusting Weights for Nonresponse

Estimating response propensities: regression models

Alternatively:

$$\hat{\phi}_i = \frac{\exp(\mathbf{x}_i\hat{\beta})}{1 + \exp(\mathbf{x}_i\hat{\beta})}$$

- **You can use the model-based $\hat{\phi}_i$ to form weight classes.**
- That is, group units with similar propensity scores into the same weighting class.
  fitted
- Then use the weight class adjustment method that we discussed.
- Pro: This allows us to easily consider many characteristics at once when constructing weighting classes.

8.6 Poststratification

- Introduced earlier as a form of ratio estimators.
- After collecting data, you separate your sample into poststrata
- We use auxiliary data of type C.
- Population sizes for poststratum h, N_h, are assumed known.

8.4 Response propensities

Auxiliary information for treating nonresponse

- This auxiliary information can come in different forms.

A. Known for every unit in the sampling frame. For example, a list of registered voters, used as sampling frame, may contain voter's age, sex, history of participating in previous elections, etc…

B. Known for the sampled units. For example, you might be interested in income but collected other information on the units like age, job title, highest level of education, etc..

C. Population means or totals from an external data source. For example, demographics from census surveys or population registry.

8.6 Poststratification

- We modify the respondent's weights so that they agree with known population count for that post stratum.

$$\omega_i^* = \omega_i \sum_{h=1}^{H} x_{ih} \frac{N_h}{\sum_{j \in \mathscr{R}} \omega_j x_{jh}}$$

← population size in stratum h

Sum of weight of respondents

- $\hat{t}_{\text{post}} = \sum_{i\mathscr{R}} \omega_i^* y_i$
- $\hat{\bar{y}}_{\text{post}} = \hat{t}_{\text{post}} / \sum_{i\mathscr{R}} \omega_i^*$

8.6 Poststratification

$$\omega_i^* = \omega_i \sum_{h=1}^{H} x_{ih} \frac{N_h}{\sum_{j \in \mathscr{R}} \omega_j x_{jh}}$$

- Why does this estimate make sense?
- What is ω_i^* within a post stratum ?

post stratum weights are recalibrated so they sum up to a know pop total.

within post stratum h, $x_{ih} = 1$

R_h : respondents in post stratum h

$$\omega_i^* = \omega_i \times \frac{N_h}{\sum_{j \in R_h} \omega_j}$$

$$\omega_i^* = \omega_i \left[\cancel{x_{11}}^{0} \frac{N_1}{\sum_{j \in R_1} \omega_j} + \cancel{x_{12}}^{0} \frac{N_2}{\sum_{j \in R_2} \omega_j} + \dots + \cancel{x_{1h}}^{1} \frac{N_h}{\sum_{j \in R_h} \omega_j} + \dots + \cancel{x_{1H}}^{0} \frac{N_H}{\sum_{j \in R_H} \omega_j} \right]$$

$$\boxed{\frac{N_h}{\ }}$$

$$\sum \omega_i^* = \frac{N_h}{\sum_{j \in R} \omega_j} \cancel{\sum_{j \in R_h} \omega_i} = N_h$$

$= \omega_i \frac{\quad}{\sum_{j \in R_h} \omega_j}$ $\rightsquigarrow$ $i \in R_h$

8.6 Poststratification

$$\omega_i^* = \omega_i \sum_{h=1}^{H} x_{ih} \frac{N_h}{\sum_{j \in \mathscr{R}} \omega_j x_{jh}}$$

- What is ω_i^* for SRS?

within post stratum h, n_{Rh}: respondents in stratum h

$$\omega_i = \frac{N}{n} \Rightarrow \omega_i^* = \frac{\cancel{N}}{\cancel{n}} \times \frac{N_h}{n_{Rh}\left(\frac{\cancel{N}}{\cancel{n}}\right)} = \boxed{\frac{N_h}{n_{Rh}}}$$

8.6 Poststratification

$$\omega_i^* = \omega_i \sum_{h=1}^{H} x_{ih} \frac{N_h}{\sum_{j \in \mathscr{R}} \omega_j x_{jh}}$$

the total estimator

- What is ~~$\hat{t}_{post}$~~ under SRS and post-stratification

$$\hat{t}_{post} = \sum_{i \in R} \omega_i^* y_i = \sum_{i \in R_1} \frac{N_1}{n_{R1}} y_i + \cdots + \sum_{i \in R_H} \frac{N_H}{n_{RH}} y_i$$

$$= N_1 \cdot \bar{y}_{R1} + \cdots + N_H \bar{y}_{RH}$$

$\bar{y}_{Rh}$: mean of respondents in stratum h

$$\hat{t}_{post} = \sum_{h=1}^{H} N_h \bar{y}_{Rh}$$

8.6 Poststratification

$$\omega_i^* = \omega_i \sum_{h=1}^{H} x_{ih} \frac{N_h}{\sum_{j \in \mathcal{R}} \omega_j x_{jh}}$$

mean estimator

- What is $\hat{\bar{y}}_{post}$ under SRS? and post-stratification

$$\hat{\bar{y}}_{post} = \frac{\hat{t}_{post}}{N} = \sum_{h=1}^{H} \left(\frac{N_h}{N} \right) \bar{y}_{Rh}$$

8.6 Poststratification

Some additional notes on post stratification

we can see that post-stratification is the same idea as before

1) Before we didn't talk about missing data but we used post-stratification to try to account for discrepancies between our sample and target population.

e.g if we sampled a smaller % of young people than in our population $\left(\frac{n_{young}}{n}\right) < \left(\frac{N_{young}}{N}\right)$

→ if younger people made less money.

→ then we weighted up our sample mean of young people $\left(\frac{N_{young}}{N}\right) \bar{y}_{young}$ so they are better represented

2) Same idea for missingness

→ so if younger age were more likely to be missing,

→ we ~~go~~ use post-stratification to upweight
the income in younger age groups
to better represent them

are better represented

8.6 Poststratification

Some additional notes on post stratification

You create post strata to deal with missingness in a similar manner to weighting classes:

At least one:

1. y_i in a stratum are all the same (or similar).

2. ϕ_i in a stratum h are all the same: $\phi_i \approx \phi_j$ in stratum h

3. y_i is uncorrelated to ϕ_i in a stratum

8.7 Calibration Methods vs Weighting Class Adjustments

Weighting class adjustments	**Calibration Methods (Poststratification/Raking)**
Weight class adjustments modify weights using information known for a selected sample.	Poststratification adjustments to weights are made using known population totals.
Can use when you have a lot of information about the non-respondent (age, sex, education…)	Can use if you don't have information about the non-respondent.

- Example: A randomized telephone survey just randomly sampling from a list of phone numbers will not have much information on the non-respondents.

- You can use post-stratification in this case.

8.6 Poststratification

Raking

- You use raking if

  1. you are using more than 1 post-stratification variable.
     - For example, you use sex (2 groups) and race (5 groups)
     - You have 10 combinations of sex and race.

AND

  2. You want to do post-stratification

BUT

8.6 Poststratification

Raking

BUT

You don't have all the population totals. E.g you don't have $\underbrace{N_{\text{male,black}}}$

But what you do have is N_{black} and N_{male} and N_{Black} , N_{White}, N_{Asian}, $N_{\text{Native American}}$, and N_{Other}

You only have the marginal totals :)

Actually quite a common thing!!!

8.6 Poststratification

Raking

How do you rake?

Consider the following sample and the weights from the sample

sum of weights in stratum Female, Black

	Black	White	Asian	Native American	Other	Sum of Weights
Female	300	1200	60	30	30	1620
Male	150	1080	90	30	30	1380
Sum of Weights	450	2280	150	60	60	3000

8.6 Poststratification

Raking

Consider that you know $N_{\text{Female}} = 1510$ and $N_{\text{Male}} = 1490$

Only look at the row data, consider calculating post strata weights ω_i^* for the two strata: Male and Female

$$\rightsquigarrow \omega_i^* = \omega_i \frac{N_h}{\sum_{i \in R_h} \omega_i} \quad [\text{stratum } h]$$

	Black	White	Asian	Native American	Other	Sum of Weights
Female	300	1200	60	30	30	1620
Male	150	1080	90	30	30	1380
Sum of Weights	450	2280	150	60	60	3000

$\Rightarrow$ stratum black, female

Female stratum

$$\omega_i^* = \omega_i \left(\frac{1510}{1620}\right)$$

$$\omega_i^* = 300 \times \left(\frac{1510}{1620}\right)$$

$$= 279.626$$

8.6 Poststratification

Raking

We know $N_{\text{Female}} = 1510$ and $N_{\text{Male}} = 1490$

We first "rake" the rows (do a post stratification adjustment) using sex

	Black	White	Asian	Native American	Other	Sum of Weights
Female	279.63	1118.52	55.93	27.96	27.96	1510
Male	161.96	1166.09	97.17	32.39	32.39	1490
Total	441.59	2284.61	153.10	60.35	60.35	3000

8.6 Poststratification

Raking

We know $N_{\text{Female}} = 1510$ and $N_{\text{Male}} = 1490$

$N_{\text{Black}} = 600$, $N_{\text{White}} = 2120$, $N_{\text{Asian}} = 150$, $N_{\text{Native American}} = 100$, and $N_{\text{Other}} = 30$

	Black	White	Asian	Native American	Other	Sum of Weights
Female	279.63	1118.52	55.93	27.96	27.96	1510
Male	161.96	1166.09	97.17	32.39	32.39	1490
Total	441.59	2284.61	153.10	60.35	60.35	3000

What about our columns? Let's rake those.

stratum black, female

$$w_i^* = w_i \left(\frac{600}{441.59}\right)$$

$$= \left(\frac{600}{\ldots}\right.$$

$= 279.63\left(\frac{}{441.59}\right)$

$= 379.94$

8.6 Poststratification

Raking

We do the same thing to the columns

$N_{\text{Black}} = 600$, $N_{\text{White}} = 2120$, $N_{\text{Asian}} = 150$, $N_{\text{Native American}} = 100$, and $N_{\text{Other}} = 30$

	Black	White	Asian	Native American	Other	Sum of Weights
Female	379.94	1037.93	54.51	46.33	13.90	1532.61
Male	220.06	1082.07	94.70	53.67	16.10	1466.60
Total	600.00	2120.00	150.00	100.00	30.00	3000

8.6 Poststratification

Raking

$N_{\text{Female}} = 1510$ and $N_{\text{Male}} = 1490$

$N_{\text{Black}} = 600$, $N_{\text{White}} = 2120$, $N_{\text{Asian}} = 150$, $N_{\text{Native American}} = 100$, and $N_{\text{Other}} = 30$

	Black	White	Asian	Native American	Other	Sum of Weights
Female	379.94	1037.93	54.51	46.33	13.90	1532.61
Male	220.06	1082.07	94.70	53.67	16.10	1466.60
Total	600.00	2120.00	150.00	100.00	30.00	3000

But now we've messed up the rows again.

8.6 Poststratification

Raking

- So you rake the rows again.
- Then…rake the columns again.
- Repeat…
- And you do this until convergence…

sum of weights in Female, black stratum

	Black	White	Asian	Native American	Other	Sum of Weights
Female	375.59	1021.47	53.72	45.56	13.67	1510
Male	224.41	1098.53	96.28	54.44	16.33	1490
Total	600.00	2120.00	150.00	100.00	30.00	3000

This was the final product after convergence for this data.

how to estimate total after?

In this case we actually don't have enough information.

we would actually need to see the raked individual weights

every individual weight would be

$w_i^* = w_i \times$ (factor adjustment from raking rows)

$\times$ (factor adjust from raking column)

$\times$ ..... etc..

currently unknown to us

If we assume this design was SRS

then $375.59 = \sum_{i \in S_{1R}} w_i^* = n_{1R} w_i^*$

since all w_i are the same. $\Rightarrow w_i^*$ are the same

S_{1R} = respondents in black, female

$\Rightarrow w_i^* = \dfrac{375.59}{n_{1R}}$

$n_{1R}: |S_{1R}|$

$\bar{y}_{1R}$: mean of respondents in black, female

$\Rightarrow \hat{t} = \sum_{i \in S_{1R}} w_i^* y_i + \sum_{i \in S_{2R}} w_i^* y_i + \ldots$

$= 375.59 \times y_{1R} + \cdots$

8.7 Imputation

- Imputation is used to assign values to the missing items
- If done right, this can reduce non-response bias.
- It also makes our dataset nice and rectangular without any holes for the missing item.

8.7 Imputation

Summary statistic imputation

- **Method:** Replace a missing value of y by a mean, median, or mode.

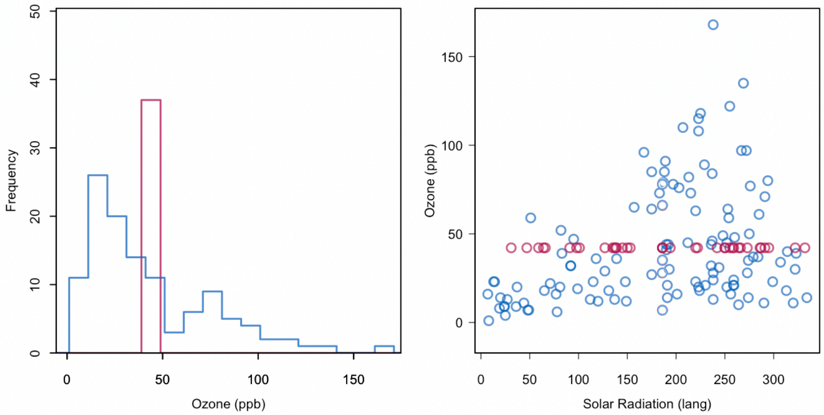


Figure 1.2: Mean imputation of `Ozone`. Blue indicates the observed data, red indicates the imputed values.

The red values are the imputed values of ozone

8.7 Imputation

Summary statistic imputation

- Pros
  - Super quick fix to missing data

- Cons:
  - Distorts distribution of variables (left figure)
  - Distorts relationship between variables (right figure)

The red values are the imputed values of ozone

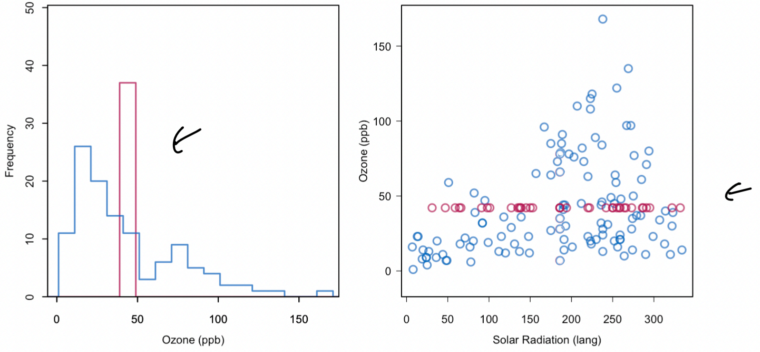


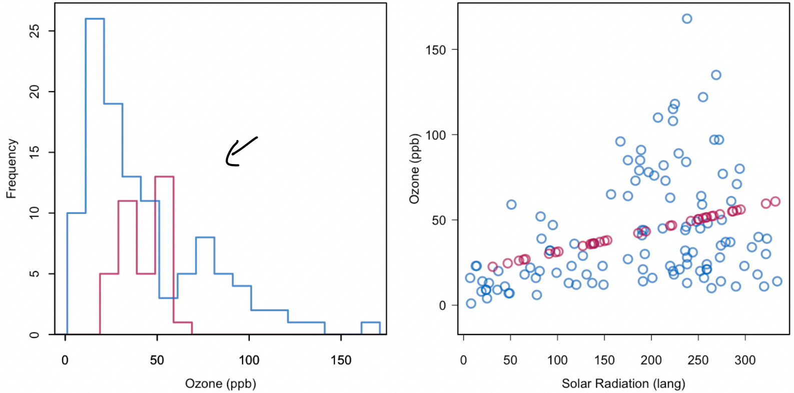
Figure 1.2: Mean imputation of `Ozone`. Blue indicates the observed data, red indicates the imputed values.

8.7 Imputation

Regression imputation

The rest of this section is out the scope of this class. It's just an FYI

- **Method:** Impute missing values with the predicted values from a linear regression model



← following regression

Figure 1.3: Regression imputation: Imputing `Ozone` from the regression line.

8.7 Imputation

Regression imputation

- **Method:** Impute missing values with the predicted values from a linear regression model
- Example: want to use $\mathbf{x}$ in the imputation model for y
- Syntax:
  1. Data for imputation model:
     - df <- data[,c("y", names(x))]
  2. Imputation step:
     - imp <- mice(df, method = "norm.predict", m=1, seed=1)

8.7 Imputation

Regression imputation

Here missing ozone values were imputed using solar radiation.

- Pros
  - Reasonable distribution of variables (left figure)
  - Average linear relationship between variables is preserved (right figure)
- Cons
  - Correlations are biased upwards. Red values have a correlation = 1 with radiation.

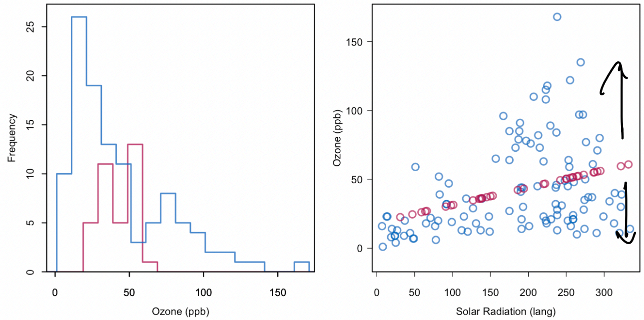


Figure 1.3: Regression imputation: Imputing `Ozone` from the regression line.

The red values are the imputed values

8.7 Imputation

Stochastic imputation

- **Stochastic imputation:** addresses the correlation bias by adding noise/uncertainty to the imputed values
- **Method:** want imputed values of y to have the same variability around the regression line as the observed values of y

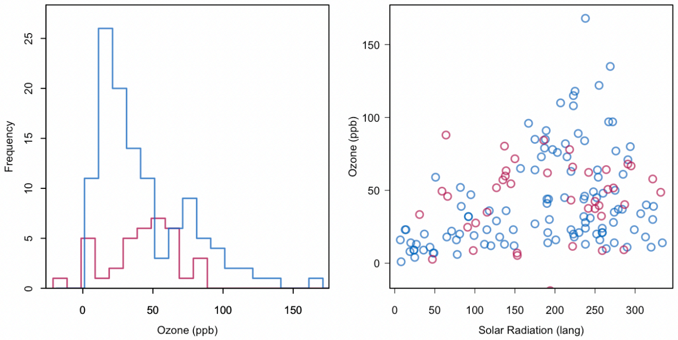


Figure 1.4: Stochastic regression imputation of `Ozone`.

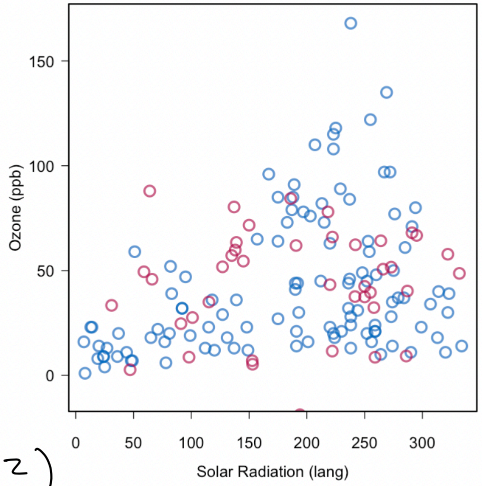
8.7 Imputation

Stochastic imputation

you get residuals from regression mode

$y - \hat{\beta}X = y - \hat{y} = \hat{\varepsilon}$

- Can add noise by:
  - Adding a random draw from the imputation model residuals to the imputed values
  - Adding random noise with SD $\hat{\sigma}$ (from the imputation model)
- Must use a seed number so the results are reproducible.



$\hat{\beta}_0$

$\hat{\beta}_1$

$\hat{\sigma}^2$

$y \sim N(\beta_0 + \beta_1 X, \sigma^2)$

simulate from $N(0, \hat{\sigma}^2) \sim \hat{\varepsilon}$

8.7 Imputation

Stochastic imputation

- **Syntax:**
  1. Data for imputation model:
     - df <- data[,c("y", names(x))]
  2. Imputation step:
     - imp <- mice(df, method = "norm.nob", m=1, seed=1)

8.7 Imputation

Stochastic imputation

Here missing ozone values were imputed using solar radiation.

- Pros:
    - All the pros of regression imputation remain
    - We have avoided inflating the correlation between variables
- Cons:
    - After imputation, we would then do this analysis on this completed data set and consider it as truth.
    - The thing is, we haven't accounted for how different our data set could look every time we rerun the imputation model.

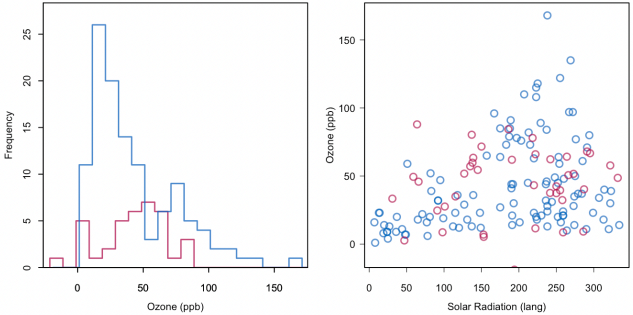


Figure 1.4: Stochastic regression imputation of `Ozone`.

8.7 Imputation

Multiple imputation

- High level idea is that we are repeating stochastic imputation multiple times.
- We end up with M imputed data sets.
- We can do our analysis M times.
- And pool together our estimates.

8.7 Imputation

Here I’m simulating a artificial population

Multiple imputation: Example

Suppose the sample was an SRS and I wanted to estimate the mean income using the sample average.

```r
library(mice)
library(survey)
library(dplyr)

set.seed(42)

# — 1. simulate a finite population survey ——————————————————
N <- 5000    # population size
n <- 300     # sample size

# simple random sample from population
pop <- data.frame(
  age    = round(rnorm(N, 40, 10)),
  educ   = sample(1:4, N, replace = TRUE)
)
pop$income <- 20000 + 500 * pop$age + 3000 * pop$educ + rnorm(N, 0, 5000)
```

```
> pop
   age educ   income
1   54    1 51467.65
2   34    2 40978.68
3   44    1 34098.57
4   46    4 55988.16
5   44    2 57612.59
6   39    4 46859.40
7   55    4 57855.08
8   39    4 56260.36
9   60    2 64220.67
10  39    3 44679.78
11  53    2 49848.63
12  63    3 51671.89
13  26    1 37075.92
14  37    4 57159.01
15  39    3 48419.83
```

8.7 Imputation

Multiple imputation: Example

```
> samp <- pop[sample(N, n), ]
> samp$weight <- N / n    # SRS weight = N/n for every unit
>
> # — 2. true population mean (what we're trying to estimate)
> true_mean <- mean(pop$income)
> cat("True population mean:", round(true_mean), "\n")
True population mean: 47359
```

- Draw an SRS of size 300
- True population mean of income is 47358.66

8.7 Imputation

Multiple imputation: Example

Simulate MAR missingness in income (missingness related to age)

```
# — 3. introduce MAR missingness on income ——————————————————
miss_prob      <- plogis(-2 + 0.15 * (40 - samp$age))
samp$income    <- ifelse(runif(n) < miss_prob, NA, samp$income)
```

← ~~rea~~ missingness depends on age.

Quite a serious case of missing values.

Mean income for observed ($n_R = 141$): 49,096.07

Mean income for non-observed ($n_M = 159$): 44,522.77

Mean income all sampled ($n = 300$): 47,266.75

8.7 Imputation

Multiple imputation: Example

Here we impute the data 500 times

```
# — 4. multiple imputation ——————————————————————————————————
# include weight as a passive variable so it rides along but isn't imputed
imp <- mice(samp, m = 500, method = "pmm", maxit = 5,
            seed = 42, printFlag = FALSE)

# — 5. fit svymean on each imputed dataset, with FPC ————————————
# svydesign needs: ids (PSUs), weights, fpc (population size or sampling fraction)
# with() passes each completed data frame in turn
fit <- with(imp, {
  des <- svydesign(
    ids     = ~1,            # SRS - no clustering
    weights = ~weight,
    fpc     = ~fpc,          # FPC column (added below)
    data    = cbind(complete(data.frame(
      age    = age,
      educ   = educ,
      income = income,
      weight = weight
    )), fpc = N)
  )
  # svymean returns an object; extract as lm-compatible via svyglm
  svyglm(income ~ 1, design = des)
})

# — 6. pool with Rubin's rules ——————————————————————————————
pooled <- pool(fit)
est    <- summary(pooled, conf.int = TRUE)
```

Here we calculate mean estimate and standard error on each ~~100~~ 500 complete data set

Here we pool the results

8.7 Imputation

Multiple imputation: Example

```
> est
         term estimate std.error statistic       df       p.value    2.5 %  97.5 % conf.low conf.high
1 (Intercept) 48193.48  494.2067  97.51686 236.0249 1.071442e-192 47219.86 49167.1 47219.86   49167.1
```

- We can see that the pooled estimate for the population mean is 48,193$ with a 95% CI [47,219.86$, 49,167.10$]
- True population mean of income is 47,358.66

Mean income for observed ($n_R = 141$): 49,096.07

Mean income for non-observed ($n_M = 159$): 44,522.77

Mean income all sampled ($n = 300$): 47,266.75

9.0 Measurement Error

Introduction

- For the remainder of this lecture, we will talk about measurement error.
- Recall, this is when the observed data is measured with error.
    - Instrument error
    - People lying
    - Recall bias
    - Too complicated of a question (how many apples did you eat over your lifetime?)

9.0 Measurement Error

Introduction

- We will briefly discuss how to reduce measurement error in the context of surveys.
- Modelling it is another topic and we leave this topic for STA302 and STA303 or more advanced modelling courses.

9.0 Measurement Error

Reducing Measurement Error

- Firstly, improve the survey quality. Here are some guidelines.

  1. Pilot test survey questions before administering the survey.

     - Verify that respondents interpret questions as intended.

     - One way to assess reliability is through **test–retest reliability**, which evaluates whether respondents provide similar answers when asked the same question at different times.

     - When using multiple questions to measure the same concept, assess the internal consistency using measures such as **Cronbach's alpha**. A higher Cronbach's alpha indicates that the questions are more consistently measuring the same underlying concept.

9.0 Measurement Error

Reducing Measurement Error

- Firstly, improve the survey quality. Here are some guidelines.

2. Write clear questions.
3. Put in procedures for administering the survey that will reduce errors.
    - E.g. Use a standardized script so every respondent being interviewed receives the same introduction and question wording.
4. Provide training and supervision for interviewers so they act consistently.
5. Give interviewers reasonable workload. They'll produce more errors if they are tired and overworked.

9.1 Sensitive Questions

- Measurement error will often happen due to the context of the survey.
- People may not want to tell the truth about a topic.
- There is a lot of evidence that suggest many people do not provide accurate answers to sensitive questions.

9.1 Sensitive Questions

Randomized Response

- Imagine the questions:
    - Have you ever shoplifted?
    - Do you use cocaine?
    - Did you understate your income on your tax returns?
- Lots of incentive for people who should answer yes to answer ‘no’.
- To try to minimize this measurement error, we are going to randomize their responses!

9.1 Sensitive Questions

Randomized Response

Note the randomized device (e.g. coin flip) can be anything but must have a known pdf

- Imagine a coin is flipped.
    - **If tails.** The subject will answer an innocent yes/no question. Eg. Is the second hand on your watch between 0 and 30.
    - **If heads.** The subject will answer the question “Do you use cocaine?”
- The person conducting the interview doesn’t know if a heads or tails was flipped.
- They just record the person’s answer.
- The person being interviewed will know the interviewer doesn’t know what question they are answering and will feel more comfortable telling the truth if they flip heads.

9.1 Sensitive Questions

Randomized Response

- How will we analyze such data?
- Suppose everyone who is asked the sensitive question tells us the truth.
- We want to estimate p_s, the proportion of responding yes to the sensitive question.

$\phi = P(\text{respondent replies yes})$

$= P(\text{responds yes} \mid \text{sensitive question})\, P(\text{sensitive question})$

$+ P(\text{responds yes} \mid \text{innocent question})\, P(\text{innocent Q})$

$= p_s \times P + p_I \times (1 - P)$

p_s: prob sens. yes $\quad$ p_I: prob. innoc. yes $\quad$ $P =$ prob sensitive question

9.1 Sensitive Questions

Randomized Response

$$\phi = P(\text{respondent replies yes})$$
$$= P(\text{respondent replies yes} \mid \text{sensisitve question})P(\text{sensitive question})$$
$$+P(\text{respondent replies yes} \mid \text{innocent question})P(\text{innocent question})$$
$$= p_S P + p_I(1 - P)$$

Let $\hat{\theta}$ be the proportion of people who answered "yes"

An estimator for p_S is:

$$\hat{\phi} = \hat{p}_s P + p_I (1-P)$$

$$\Rightarrow \boxed{\hat{p}_s = \frac{\hat{\phi} - p_I(1-P)}{P}}$$

The estimated variance for $\hat{p}_S$:

$$Var(\hat{p}_s) = \frac{Var(\hat{\theta})}{P^2} \qquad Var(\hat{p}_s) = \frac{\hat{\theta}(1-\hat{\theta})}{nP^2}$$

9.1 Sensitive Questions

Randomized Response

Example:

An SRS of high school seniors is selected. Each senior in the sample is presented with a card containing the following two questions.

A. Have you ever cheated on an exam?

B. Were you born in July?

- We know from birth records $p_I = 0.085$.
- Suppose the randomized device had $P = 1/5$
- Of 800 people interviewed, 175 said yes.

Estimate the proportion of students who ever cheated on an exam.

$$\hat{p}_S = \frac{\left(\frac{175}{800}\right) - 0.085\left(1-\frac{1}{5}\right)}{\frac{1}{5}}$$

$$= 0.75375$$

$\Rightarrow$ est. 75.4% of students cheated on exam

9.1 Sensitive Questions

Randomized Response

There's other types of randomization we can do.

Example:

An SRS of high school seniors is selected.

A. Flip Tails, answer question: Have you ever cheated on an exam?

B. Flip Heads, answer yes.

$P_I = 1$

The instructor doesn't see the coin.
Suppose 70% of people answered 'yes'

Estimate the proportion of students who ever cheated on an exam.

$$\hat{p}_s = \frac{\hat{\phi} - 1(0.5)}{0.5}$$

$$= \frac{0.7 - 0.5}{0.5}$$

$$= 40\%$$

Progress

- ☑ Describe common sources of nonresponse and measurement error in surveys.
- ☑ Explain how nonresponse and measurement error can affect the accuracy of survey estimates.
- ☑ Construct nonresponse-adjusted survey weights using weighting class adjustments, post-stratification, and raking.
- ☑ Explain how auxiliary information can be used to reduce nonresponse bias.
- ☑ Describe the purpose of imputation and explain how imputation can be used to address missing data.
- ☑ Describe common strategies for reducing measurement error in survey design collection.

Thanks for the great semester! Have a great summer!

If you have time, fill out the course evaluation form.

First time developing this course and any constructive feedback is welcome.

Survey Sam & Obs Data STA304H1-F-LEC0101

Students

https://go.blueja.io/mZZ6LU4YWU6SOS5_tYY4tQ